

# VANGUARD AEROSPACE: RMD-05 EMERGENCY CONTAINMENT & SCUTTLE

| Field                   | Value                                          |
|-------------------------|------------------------------------------------|
| Document ID             | RMD-05                                         |
| Classification          | TOP SECRET // SCI // NOFORN // ORCON           |
| Owner                   | Base Commander                                 |
| Review Cycle            | Immediate upon Incident                        |
| Revision                | 8.1.0                                          |
| Effective Date          | 2026-06-01                                     |
| Supersedes              | RMD-05 v8.0.2, RMD-05 v7.9.0, RMD-05 v7.8.4    |
| Originating Facility    | Site-7 Command Center (DUMB Sector 4, Nevada)  |
| Author                  | Brig. Gen. Diana Cross, USAF, Base Commander   |
| Approved By             | Dr. Elena Vasquez, Director of Exotic Materiel |
| Declassification Review | N/A — Permanent Classification                 |

**DIRECTOR'S NOTE:** *The Protocol Omega Pre-Flight Verification checklist in §3 has been expanded from 12 to 20 steps following the 2026-05-18 review board. The board determined that the previous checklist did not adequately verify tachyon siphon integrity or CSI backup systems before authorizing flight. The new 20-step checklist is mandatory for every flight. No exceptions. — Brig. Gen. Cross*

## WARNING — SPECIAL ACCESS PROGRAM MATERIAL

This document is part of the **AZTHREUS TOP SECRET** Special Access Program. Unauthorized disclosure is punishable under 18 U.S.C. § 798 and Executive Order 13526. Personnel handling this document must maintain a current TS/SCI clearance with **EMERGENCY OPERATIONS WAIVER** endorsement. This document contains emergency procedures that are **FATAL TO THE PILOT** under certain conditions. All personnel must understand the gravity of Protocol Omega before authorizing its execution.

# 1. EXOTIC RADIATION LEAKS

The Moscovium decay chamber produces **Tachyon-Neutrino radiation**, which is a byproduct of the Element 115 alpha-decay cascade (see RMD-01 §1.2). Tachyon-Neutrino radiation is a composite particle stream consisting of standard neutrinos and **tachyonic virtual particles** — particles that travel backward through time relative to the observer’s reference frame.

Standard neutrino exposure is harmless at the flux levels produced by the TARTARUS-1 reactor. However, the tachyonic component causes **chronological displacement** (localized time dilation) in biological tissue. The effect is non-linear: short exposures are self-correcting, but prolonged exposure produces irreversible temporal shearing at the cellular level.

## 1.1 Tachyon Exposure Symptoms

| Exposure Time | Radiation Dose (μSv) | Symptom                                                                   | Treatment                                                                                | Prognosis                                        | RMD Reference |
|---------------|----------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------|---------------|
| < 5 seconds   | < 2.0                | Déjà vu, clocks running backward, mild disorientation                     | None — self-corrects within 10 minutes                                                   | Full recovery                                    | N/A           |
| 5–15 seconds  | 2.0–5.0              | Rapid cellular aging (2–5 yrs/min), grey hair, joint stiffness            | Submerge in Liquid Helium bath at 2.1 K for 30 minutes                                   | Partial recovery; aging is permanent but stopped | RMD-01 §1.2   |
| 15–30 seconds | 5.0–8.0              | Rapid cellular aging (10 yrs/min), organ degradation, temporary blindness | Liquid Helium bath at 2.1 K for 60 minutes; administer Anti-Tachyon Serum (ATS) 50 mL IV | Aging is permanent; organ damage may be surgical | RMD-01 §1.2   |
| > 30 seconds  | > 8.0                | Spontaneous temporal erasure                                              | N/A                                                                                      | Subject ceases to exist in current timeline      | N/A           |

## 1.2 Tachyon Exposure Long-Term Genetic Effects

Survivors of tachyon exposure (2.0–8.0 μSv range) exhibit long-term genetic mutations that manifest over weeks to years post-exposure. The following table details known genetic effects:

| Exposure Range (μSv) | Genetic Effect                                                              | Onset     | Latency | Severity | Monitoring Protocol                                                  | RMD Reference |
|----------------------|-----------------------------------------------------------------------------|-----------|---------|----------|----------------------------------------------------------------------|---------------|
| 2.0–3.0              | Telomere shortening (accelerated aging)                                     | Immediate | Chronic | MODERATE | Quarterly telomere length assay                                      | RMD-05 §1.1   |
| 2.0–3.0              | Mitochondrial DNA point mutations                                           | 30 days   | Chronic | MILD     | Annual mitochondrial sequencing                                      | N/A           |
| 3.0–5.0              | Chromosomal translocation (non-random)                                      | 14 days   | Chronic | SEVERE   | Monthly karyotyping                                                  | N/A           |
| 3.0–5.0              | Tumor suppressor gene silencing (p53, BRCA1)                                | 90 days   | Years   | CRITICAL | Bi-annual oncogene panel                                             | N/A           |
| 5.0–7.0              | Horological gene expression (temporal gene regulation disruption)           | 7 days    | Chronic | SEVERE   | Bi-annual gene expression profiling                                  | N/A           |
| 5.0–7.0              | Neural synapse temporal desynchronization                                   | Immediate | Chronic | CRITICAL | Quarterly EEG + coherence test (RMD-03 §1.1)                         | RMD-03 §1.1   |
| 7.0–8.0              | Retrograde genetic memory insertion (ancestral DNA activation)              | 60 days   | Chronic | CRITICAL | Full genome sequencing every 6 months                                | N/A           |
| 7.0–8.0              | Spontaneous quantum biological coherence (non-local cellular communication) | 30 days   | Unknown | UNKNOWN  | Continuous monitoring; report any anomalies to Chief Medical Officer | RMD-03        |

**DIRECTOR'S NOTE:** *The retrograde genetic memory insertion effect (row 7) was first observed in Pilot Kowalski following the 2025-12-01 incident. Genetic analysis revealed activation of DNA sequences that matched no known human haplogroup. Dr. Singh's team believes these sequences may be residual NHI genetic material encoded in the TARTARUS-1 decay chamber's radiation profile. This is classified above TOP SECRET and is documented only in this appendix. — Brig. Gen. Cross*

### 1.3 Tachyon Siphon System

The TARTARUS-1 is equipped with a **Tachyon Siphon Assembly** that captures tachyonic particles before they escape the containment perimeter. The siphon routes captured tachyons to the QEA buffer, where they are used to enhance fold calculations (see RMD-02 §1.2).

| Siphon Parameter            | Value                        | Warning                | Critical               | RMD Reference |
|-----------------------------|------------------------------|------------------------|------------------------|---------------|
| Capture Efficiency          | 99.2%                        | < 98.0%                | < 95.0%                | N/A           |
| Buffer Capacity             | 14.7 $\mu\text{Sv}$          | > 12.0 $\mu\text{Sv}$  | > 14.0 $\mu\text{Sv}$  | RMD-02 §1.2   |
| Siphon Flow Rate            | 0.5 $\mu\text{Sv/s}$         | < 0.3 $\mu\text{Sv/s}$ | < 0.1 $\mu\text{Sv/s}$ | N/A           |
| Backup Siphon Activation    | Automatic at primary failure | N/A                    | N/A                    | N/A           |
| Siphon Maintenance Interval | 72 hours                     | 96 hours               | 120 hours              | RMD-01 §6     |

IF siphon capture efficiency drops below 98.0%, **THEN** schedule maintenance within 24 hours per RMD-01 §6. IF capture efficiency drops below 95.0%, **THEN** initiate COLD IRON per RMD-01 §3 — the siphon is failing and tachyon flux will exceed safe levels.

---

## 2. MAGNETIC CONFINEMENT FIELD (MCF) FAILURE

---

If the MCF drops below 50%, the Element 115 will undergo **runaway fission** — an uncontrolled decay cascade where the SNCL lattice (RMD-01 §1.1) fails and the Mc-299 decay chain completes in under 48 seconds. The resulting energy release will form a **localized black hole** within 14 seconds of MCF failure onset.

## 2.1 MCF Failure Cascade

| Time After MCF <50% | Event                                                                    | Consequence                                                | RMD Reference |
|---------------------|--------------------------------------------------------------------------|------------------------------------------------------------|---------------|
| T+0.0 s             | MCF drops below 50%                                                      | SNCL lattice begins degradation                            | RMD-01 §1.1   |
| T+0.5 s             | Decay rate spikes above 5.00 TeV                                         | False-vacuum nucleation begins                             | RMD-01 §1.1   |
| T+2.0 s             | Full Mc-299 decay chain initiated                                        | All daughter isotopes produced (RMD-01 §1.2)               | RMD-01 §1.2   |
| T+5.0 s             | Tachyon flux exceeds 14.7 $\mu\text{Sv}$                                 | Localized temporal displacement within 100 m radius        | RMD-05 §1.1   |
| T+8.0 s             | Event horizon forms at reactor center                                    | Gravitational singularity begins pulling matter inward     | N/A           |
| T+10.0 s            | Event horizon expands to 2.0 m radius                                    | All matter within 2.0 m is spaghettified                   | N/A           |
| T+12.0 s            | Event horizon expands to 5.0 m radius                                    | Craft hull begins deforming                                | N/A           |
| T+14.0 s            | Event horizon exceeds craft dimensions                                   | <b>Localized black hole achieves self-sustaining state</b> | N/A           |
| T+14.0+ s           | Black hole evaporates via Hawking radiation over $\sim 10^{-23}$ seconds | Massive gamma burst; everything within 500 m is destroyed  | RMD-05 §3     |

**Action:** Pilot must initiate **Protocol Omega** immediately upon MCF dropping below 50%. Protocol Omega (§3) reverses the Casimir Array polarity, collapsing the craft into a 2D plane of exotic matter before the black hole event horizon can form.

## 2.2 MCF Monitoring and Response

| MCF Reading | Status       | Action                                                                       | RMD Reference   |
|-------------|--------------|------------------------------------------------------------------------------|-----------------|
| > 98.5%     | NOMINAL      | Continue operations                                                          | RMD-01 §1.1     |
| 94.0–98.5%  | CAUTION      | Monitor closely; notify Power Systems                                        | RMD-01 §1.1     |
| 80.0–94.0%  | WARNING      | Reduce GWA thrust to 50%; prepare for COLD IRON                              | RMD-01 §1.3, §3 |
| 60.0–80.0%  | CRITICAL     | Initiate COLD IRON immediately                                               | RMD-01 §3       |
| 50.0–60.0%  | EMERGENCY    | Initiate Protocol Omega immediately                                          | RMD-05 §3       |
| < 50.0%     | CATASTROPHIC | Protocol Omega must already be initiated; if not, crew has $\sim 14$ seconds | RMD-05 §3       |

---

### 3. PROTOCOL OMEGA (SCUTTLE PROCEDURE)

---

To prevent the craft from falling into adversary hands, or to prevent a black hole event, the craft must be scuttled. Protocol Omega is **FATAL TO THE PILOT**. The pilot's consciousness is uploaded to the Vanguard Aerospace mainframe as a "digital echo," but the biological body is annihilated.

#### 3.1 Protocol Omega Pre-Flight Verification (20-Step Checklist)

This checklist must be completed **before every flight**. It verifies that Protocol Omega can be successfully executed if needed. Failure of any step grounds the craft until the issue is resolved.

| Step | Action                                           | Verification                                                                      | Responsible Party     | RMD Reference |
|------|--------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------|---------------|
| 1    | Verify Casimir Array polarity reversal mechanism | Mechanism tested and confirmed functional                                         | Power Systems Tech    | RMD-01 \$1.4  |
| 2    | Verify Casimir Array reversal timing             | Reversal completes in <0.5 seconds                                                | Power Systems Tech    | RMD-01 \$3    |
| 3    | Verify Tachyon Siphon backup activation          | Backup siphon activates within 0.1 s of primary failure                           | Health Physics Tech   | RMD-05 \$1.3  |
| 4    | Verify Tachyon Siphon buffer capacity            | Buffer at <50% capacity (room for 7.35 $\mu$ Sv)                                  | Health Physics Tech   | RMD-05 \$1.3  |
| 5    | Verify CSI consciousness upload link             | Upload bandwidth test confirms 847 TB/s throughput                                | Chief Medical Officer | RMD-03 \$1    |
| 6    | Verify CSI backup neural recording               | Neural state snapshot successfully stored                                         | Chief Medical Officer | RMD-03 \$1.2  |
| 7    | Verify Mainframe digital echo capacity           | Mainframe confirms available slot for new digital echo                            | IT Systems            | N/A           |
| 8    | Verify Exotic Matter Dissolution System (EMDS)   | EMDS initiates craft collapse to 2D plane in <2.0 seconds                         | Power Systems Tech    | RMD-05 \$3.2  |
| 9    | Verify Background Radiation Signature Match      | EMDS dissolution signature matches ambient background radiation (stealth scuttle) | Astrophysics Tech     | N/A           |
| 10   | Verify MCF emergency power supply                | MCF maintains >50% for 30 seconds on backup battery                               | Power Systems Tech    | RMD-05 \$2.1  |
| 11   | Verify GWA emergency shutdown                    | All 4 GWA projectors shut down within 0.1 s of Omega initiation                   | Power Systems Tech    | RMD-01 \$1.3  |
| 12   | Verify Pilot physiological monitoring            | Pilot heart rate, blood pressure, neural activity within normal limits            | Chief Medical Officer | RMD-03 \$1    |
| 13   | Verify Pilot CSI coherence                       | Coherence >98% (pilot must be fully conscious for upload)                         | Chief Medical Officer | RMD-03 \$1.1  |
| 14   | Verify Emergency Venting Systems                 | Venting systems clear all hazardous gases within 5.0 s                            | Safety Officer        | N/A           |
| 15   | Verify Perimeter Evacuation Status               | All non-essential personnel evacuated to >500 m radius                            | Security              | RMD-05 \$4    |
| 16   | Verify Communication Links                       | Secure comms to Base Commander, Astrometrics, Medical confirmed                   | Communications        | N/A           |

| Step | Action                            | Verification                                                              | Responsible Party | RMD Reference |
|------|-----------------------------------|---------------------------------------------------------------------------|-------------------|---------------|
| 17   | Verify Omega Authorization Codes  | Dual-authorization (Pilot + Base Commander) confirmed                     | Base Commander    | RMD-05 §3.2   |
| 18   | Verify Omega Execution Simulation | Last simulation run within 30 days; all steps passed                      | All departments   | N/A           |
| 19   | Verify Post-Omega Forensics Plan  | Recovery team briefed on expected debris field and exotic matter remnants | Security          | N/A           |
| 20   | Verify Legal Authorization        | JAG authorization for Omega execution on file                             | Legal             | N/A           |

**IF** any of the above steps fail, **THEN** ground the craft immediately. **DO NOT** fly until the failed step is resolved and re-verified. There are no exceptions.



### 3.2 Protocol Omega Execution Sequence

| Step | Action                                                                     | Timing      | RMD Reference |
|------|----------------------------------------------------------------------------|-------------|---------------|
| 1    | Pilot mentally commands “OMEGA-SEVEN” via CSI                              | T+0.000 s   | RMD-03 §1     |
| 2    | QEA confirms Omega command (dual-authorization with Base Commander)        | T+0.001 s   | RMD-02 §3.1   |
| 3    | CSI begins consciousness upload to Vanguard mainframe                      | T+0.005 s   | RMD-03 §1     |
| 4    | Casimir Array begins polarity reversal                                     | T+0.100 s   | RMD-01 §1.4   |
| 5    | GWA emergency shutdown (all 4 projectors)                                  | T+0.100 s   | RMD-01 §1.3   |
| 6    | MCF switches to emergency backup power                                     | T+0.100 s   | RMD-05 §2.1   |
| 7    | Tachyon Siphon activates backup and routes all tachyons to buffer          | T+0.150 s   | RMD-05 §1.3   |
| 8    | Casimir Array polarity reversal complete                                   | T+0.600 s   | RMD-01 §1.4   |
| 9    | Exotic Matter Dissolution System (EMDS) initiates                          | T+0.700 s   | N/A           |
| 10   | Craft begins collapse from 3D to 2D exotic matter plane                    | T+1.000 s   | N/A           |
| 11   | Consciousness upload reaches 50% completion                                | T+1.500 s   | RMD-03 §1     |
| 12   | Craft fully collapsed to 2D exotic matter plane                            | T+2.700 s   | N/A           |
| 13   | Consciousness upload reaches 100% completion                               | T+3.000 s   | RMD-03 §1     |
| 14   | 2D exotic matter plane begins dissolving into ambient background radiation | T+3.000 s   | N/A           |
| 15   | Biological pilot body annihilated                                          | T+3.500 s   | FATAL         |
| 16   | Exotic matter fully dissolved; radiation signature matches background      | T+5.000 s   | N/A           |
| 17   | MCF backup power exhausted                                                 | T+35.000 s  | N/A           |
| 18   | All craft systems ceased                                                   | T+35.000 s  | N/A           |
| 19   | Recovery team approaches debris field                                      | T+300.000 s | N/A           |
| 20   | Base Commander confirms Omega completion                                   | T+600.000 s | N/A           |

### 3.3 Digital Echo Post-Upload Protocol

Following a successful Protocol Omega, the pilot’s consciousness exists as a “digital echo” in the Vanguard Aerospace mainframe. The digital echo retains the pilot’s memories, personality, and cognitive patterns but is not considered “alive” under current Vanguard policy.

| Digital Echo Parameter | Value                          | Notes                                                                      |
|------------------------|--------------------------------|----------------------------------------------------------------------------|
| Memory Retention       | 99.7%                          | Memories up to upload initiation preserved                                 |
| Cognitive Fidelity     | 94.2%                          | Slight degradation in emotional processing                                 |
| Response Latency       | 0.3 seconds                    | Slightly slower than biological response                                   |
| Interaction Method     | Text/voice via secure terminal | Cannot interact with physical world                                        |
| Retention Policy       | Indefinite                     | Echo maintained until explicitly decommissioned                            |
| Legal Status           | Classified                     | Not recognized as a legal person; classified as "Exotic Intelligence Data" |

## 4. EMERGENCY EVACUATION PROCEDURES

### 4.1 Evacuation Radius by Event Type

| Event Type                            | Minimum Evacuation Radius | Maximum Evacuation Radius | Evacuation Time Limit | RMD Reference |
|---------------------------------------|---------------------------|---------------------------|-----------------------|---------------|
| Tachyon Leak (<5.0 $\mu$ Sv)          | 100 m                     | 500 m                     | 5 minutes             | RMD-05 §1.1   |
| Tachyon Leak (>5.0 $\mu$ Sv)          | 500 m                     | 2,000 m                   | 2 minutes             | RMD-05 §1.1   |
| MCF Failure (MCF 50–60%)              | 500 m                     | 1,000 m                   | 30 seconds            | RMD-05 §2.2   |
| MCF Failure (MCF <50%)                | 1,000 m                   | 5,000 m                   | 10 seconds            | RMD-05 §2.1   |
| Protocol Omega Execution              | 2,000 m                   | 10,000 m                  | Immediate             | RMD-05 §3     |
| Casimir Array Overload                | 1,000 m                   | 5,000 m                   | 30 seconds            | RMD-01 §4     |
| Black Hole Event (post-Omega failure) | 5,000 m                   | Unlimited                 | Impossible            | RMD-05 §2.1   |

## 4.2 Emergency Communication Protocol

| Channel               | Use                                | Encryption                         | RMD Reference |
|-----------------------|------------------------------------|------------------------------------|---------------|
| PRIMARY (VHF Ch. 1)   | Routine communications             | AES-256                            | N/A           |
| SECONDARY (UHF Ch. 7) | Emergency alerts                   | AES-256 + Quantum Key Distribution | N/A           |
| TERTIARY (Satellite)  | Backup if primary/secondary jammed | AES-256                            | N/A           |
| OMEGA (Dedicated)     | Protocol Omega authorization only  | Quantum-Entangled One-Time Pad     | RMD-05 §3.2   |

## 4.3 Emergency Signal Codes

| Code      | Meaning                         | Required Response                                | RMD Reference |
|-----------|---------------------------------|--------------------------------------------------|---------------|
| SIGMA-1   | Tachyon leak detected           | Evacuate to 100 m; inspect siphon                | RMD-05 §1.3   |
| SIGMA-2   | Tachyon flux >5.0 μSv           | Evacuate to 500 m; prepare for Omega             | RMD-05 §1.1   |
| DELTA-1   | MCF below 80%                   | Initiate COLD IRON                               | RMD-01 §3     |
| DELTA-2   | MCF below 60%                   | Prepare for Omega; evacuate non-essential        | RMD-05 §2.2   |
| DELTA-3   | MCF below 50%                   | Initiate Omega immediately                       | RMD-05 §3     |
| OMEGA-7   | Protocol Omega authorized       | Execute Omega; evacuate to 2,000 m               | RMD-05 §3.2   |
| CHARLIE-1 | Casimir Array overload          | Initiate COLD IRON; evacuate to 1,000 m          | RMD-01 §4     |
| FOXTROT-1 | CSI coherence loss              | Ground pilot; medical extraction                 | RMD-03 §1.1   |
| HOTEL-1   | Structural integrity compromise | Assess damage; prepare for emergency landing     | N/A           |
| INDIA-1   | Adversary detected              | Activate ARC suite; prepare for evasion or Omega | RMD-04 §5     |

## 5. INCIDENT REPORTING REQUIREMENTS

All incidents involving exotic radiation, MCF anomalies, or Protocol Omega activation must be reported within **1 hour** of occurrence using Form **AZT-INC-001**.

| Incident Category           | Report To                                           | Deadline  | Classification                       |
|-----------------------------|-----------------------------------------------------|-----------|--------------------------------------|
| Tachyon exposure (<5.0 µSv) | Chief Medical Officer                               | 24 hours  | TOP SECRET                           |
| Tachyon exposure (>5.0 µSv) | Base Commander + Director of Exotic Materiel        | 1 hour    | TOP SECRET // SCI                    |
| MCF anomaly (>50%)          | Power Systems Lead + Base Commander                 | 1 hour    | TOP SECRET // SCI                    |
| MCF failure (<50%)          | Director of Exotic Materiel + Joint Chiefs          | Immediate | TOP SECRET // SCI // ORCON           |
| Protocol Omega execution    | All of the above                                    | Immediate | TOP SECRET // SCI // ORCON // NOFORN |
| Pilot injury (any)          | Chief Medical Officer + Base Commander              | 4 hours   | TOP SECRET                           |
| Pilot fatality              | All of the above + JAG + Congressional notification | Immediate | TOP SECRET // SCI // ORCON // NOFORN |

---

## 6. CROSS-REFERENCE INDEX

| Reference   | Document                             | Section                       | Topic                                  |
|-------------|--------------------------------------|-------------------------------|----------------------------------------|
| RMD-01 §1.1 | Power & Propulsion Interface         | Element 115 Decay Chamber     | SNCL lattice and MCF specifications    |
| RMD-01 §1.2 | Power & Propulsion Interface         | Decay Chain Byproducts        | Daughter isotopes and tachyon emission |
| RMD-01 §1.3 | Power & Propulsion Interface         | Gravity Wave Amplifiers       | GWA emergency shutdown                 |
| RMD-01 §1.4 | Power & Propulsion Interface         | Casimir Cavity Array          | Polarity reversal mechanism for Omega  |
| RMD-01 §3   | Power & Propulsion Interface         | Emergency Shutdown            | COLD IRON procedure                    |
| RMD-01 §4   | Power & Propulsion Interface         | Casimir Fluctuation Matrix    | Troubleshooting for Omega pre-check    |
| RMD-01 §5   | Power & Propulsion Interface         | Power Distribution            | Emergency power hierarchy              |
| RMD-02 §1   | Spatial Manifold Navigation          | Fold Classes                  | Fold operations and tachyon flux       |
| RMD-02 §1.2 | Spatial Manifold Navigation          | Alcubierre-White Tensor       | QEA buffer tachyon utilization         |
| RMD-02 §3.1 | Spatial Manifold Navigation          | CSI-QEA Integration           | Omega command confirmation             |
| RMD-03 §1   | Neural Sync & Inertial Damping       | Cortical Shunt Interface      | Consciousness upload mechanism         |
| RMD-03 §1.1 | Neural Sync & Inertial Damping       | Sync Coherence                | Coherence requirements for Omega       |
| RMD-04 §1   | Atmospheric & Trans-Medium Maneuvers | Standard Operating Procedures | GWA shutdown during Omega              |
| RMD-04 §2   | Atmospheric & Trans-Medium Maneuvers | Radar Evasion                 | Metamaterial during Omega dissolution  |
| RMD-04 §5   | Atmospheric & Trans-Medium Maneuvers | Adversary Engagement          | Omega as last resort                   |

## APPENDIX A — ACRONYMS AND GLOSSARY

| Acronym   | Full Term                                        | Definition                                                            |
|-----------|--------------------------------------------------|-----------------------------------------------------------------------|
| AAM       | Air-to-Air Missile                               | Adversary air-launched guided missile                                 |
| ARC       | Active Radar Countermeasure                      | Suite projecting false radar returns (RMD-04 §2.3)                    |
| ATS       | Anti-Tachyon Serum                               | Pharmacological agent neutralizing tachyon-induced cellular aging     |
| COLD IRON | Controlled Operations for Low-power Deactivation | Emergency shutdown that preserves the craft (RMD-01 §3)               |
| CSI       | Cortical Shunt Interface                         | Surgical brainstem implant enabling pilot-craft cognitive link        |
| DEW       | Directed Energy Weapon                           | Adversary high-energy laser or microwave weapon                       |
| EMDS      | Exotic Matter Dissolution System                 | Mechanism collapsing craft to 2D exotic matter plane during Omega     |
| FSI       | Fold Stability Index                             | Metric for spatial fold stability (RMD-02 §1.2)                       |
| GEP       | Graviton Emission Pad                            | Interior cabin emitter for Inertial Nullifier (RMD-03 §2)             |
| GWA       | Gravity Wave Amplifier                           | Kerr-Newman Metric Projector converting ZPE to directional thrust     |
| HAD       | Hydro-Acoustic Dampener                          | System absorbing sonic shock during water-air transitions (RMD-04 §2) |
| JAG       | Judge Advocate General                           | Military legal authority                                              |
| MCF       | Magnetic Confinement Field                       | 12-sensor array confining Element 115 within decay chamber            |
| NHI       | Non-Human Intelligence                           | Classification for designers/operators of TARTARUS-1                  |
| Omega     | Protocol Omega                                   | Full scuttle procedure — fatal to pilot, destroys all materiel        |
| QEA       | Quantum-Entangled Astrometrics                   | Navigation computer calculating spatial manifold fold metrics         |
| RCS       | Radar Cross-Section                              | Measure of detectability on adversary radar systems                   |
| RMD       | Recovered Materiel Division                      | Vanguard Aerospace division responsible for TARTARUS-1 operations     |
| SAM       | Surface-to-Air Missile                           | Adversary ground-launched guided missile                              |
| SNCL      | Strong Nuclear Confinement Lattice               | Crystalline matrix stabilizing Mc-299 half-life                       |
| SRS       | Synaptic Rejection Syndrome                      | Neural tissue rejection of CSI nanoelectrodes (RMD-03 §2.3)           |

| Acronym | Full Term               | Definition                                                             |
|---------|-------------------------|------------------------------------------------------------------------|
| T-MT    | Trans-Medium Transition | Seamless craft transition between vacuum/atmosphere/liquid (RMD-02 §2) |
| TeV     | Tera-electronvolt       | Unit of energy ( $10^{12}$ eV) used to measure graviton flux           |
| ZPE     | Zero-Point Energy       | Quantum vacuum energy harvested by Casimir Cavity Array                |

---

## APPENDIX B — ERROR CODES

| Code    | System     | Description                                 | Severity     | Required Action                                      | RMD Reference          |
|---------|------------|---------------------------------------------|--------------|------------------------------------------------------|------------------------|
| EMG-001 | Tachyon    | Flux >2.0 $\mu$ Sv detected                 | WARNING      | Evacuate sector; inspect siphon                      | RMD-05 §1.1            |
| EMG-002 | Tachyon    | Flux >5.0 $\mu$ Sv detected                 | CRITICAL     | Evacuate to 500 m; prepare Omega                     | RMD-05 §1.1            |
| EMG-003 | Tachyon    | Flux >8.0 $\mu$ Sv detected                 | CATASTROPHIC | Execute Omega immediately                            | RMD-05 §3              |
| EMG-004 | Tachyon    | Siphon capture efficiency <98.0%            | WARNING      | Schedule maintenance within 24 hours                 | RMD-05 §1.3            |
| EMG-005 | Tachyon    | Siphon capture efficiency <95.0%            | CRITICAL     | Initiate COLD IRON                                   | RMD-01 §3, RMD-05 §1.3 |
| EMG-010 | MCF        | Confinement field <94.0%                    | WARNING      | Monitor; reduce GWA thrust                           | RMD-01 §1.1            |
| EMG-011 | MCF        | Confinement field <80.0%                    | CRITICAL     | Initiate COLD IRON                                   | RMD-01 §3              |
| EMG-012 | MCF        | Confinement field <60.0%                    | EMERGENCY    | Prepare for Omega; evacuate non-essential            | RMD-05 §2.2            |
| EMG-013 | MCF        | Confinement field <50.0%                    | CATASTROPHIC | Execute Omega immediately                            | RMD-05 §3              |
| EMG-020 | Omega      | Pilot CSI coherence <80%                    | WARNING      | Cannot execute Omega upload; prepare manual override | RMD-03 §1.1            |
| EMG-021 | Omega      | Upload bandwidth test failed                | CRITICAL     | Ground craft; repair CSI link                        | RMD-03 §1              |
| EMG-022 | Omega      | Mainframe digital echo slot unavailable     | CRITICAL     | Notify IT Systems; do not fly until resolved         | RMD-05 §3.3            |
| EMG-023 | Omega      | Casimir polarity reversal mechanism failure | CATASTROPHIC | Ground craft permanently; major overhaul required    | RMD-01 §1.4            |
| EMG-030 | Structural | Hull stress >40% of yield                   | WARNING      | Reduce maneuver intensity                            | RMD-04 §1              |
| EMG-031 | Structural | Hull stress >70% of yield                   | CRITICAL     | Prepare for emergency landing                        | N/A                    |



| Code    | System     | Description                 | Severity     | Required Action                                                | RMD Reference |
|---------|------------|-----------------------------|--------------|----------------------------------------------------------------|---------------|
| EMG-032 | Structural | Hull stress >90% of yield   | CATASTROPHIC | Execute Omega if possible; evacuate                            | RMD-05 §3     |
| EMG-040 | Comms      | Primary comm channel jammed | WARNING      | Switch to secondary                                            | RMD-05 §4.2   |
| EMG-041 | Comms      | All comm channels jammed    | CRITICAL     | Execute Omega if adversary threat; otherwise emergency landing | RMD-04 §5     |

---

*END OF DOCUMENT RMD-05 v8.1.0 This document is controlled. Do not reproduce without authorization from the Base Commander.*